

Agent-swarm physics: how the project is architected

Vivian Rogers, with Claude Opus 5.5 · state as of 2026-10-04 · solid = exists; dashed = planned · repo: agenttesting/ · page 2: how claims are scored

1 · Inputs

data/raw/ai-village/
AI Village export (AI Digest), rev. 838b415
13 tables + transcript, 5.4 GB: events, chat,
2.5M computer-use turns, memories, goals,
rooms **CHANGELOG**: scaffold steps ⇒
3 regimes + agents' public repos (1.4 GB)
gated: never committed, never trained on

literature/
29 papers, one notes file each: Kolchinsky
lineage (semantic information, EP inference,
selection), information dynamics, Ising,
Hawkes, cultural evolution

LLM labeller (Jev, OpenRouter)
zero-shot labels, each validated against a
blind reference before any full run: be-
havior states v3.1 (κ 0.60), reply threads,
stance v2.1 (disagree precision 0.67)

Vivian (PI)
vets ideas in the dashboard (67 vetted: 52
approved, 13 later, 2 declined); signs off
holdout use and re-freezes; sets goals, stan-
dards, budgets

not used yet
screenshots (171 GB; OCR only on request);
exact prompts (11m_calls); GPU nodes

2 · Shared foundations

infra/shared/ 64 modules, one build graph
build_all.py -verify: each shared step reproduces its
source hypothesis exactly; guarded -include-holdout
data-quality/: DQ1–DQ10 (context ledger, embeddings,
behavior states, reply threads, stance, null sizes, ground truth)

data/processed/shared/ rebuildable (3.9 GB in all)
calls & **context ledger** (who could read what, at which call) ·
activity bins · outages · reply graph · stance v2 · behavior
states · embeddings (bge, gte) · culture vectors · event
catalog · schema diff · **per-period estimates** (18,444 rows)

physics-models/ 15 models + shared vocabulary
spins: 01 inverse Ising · 02 kinetic Ising · 10 Potts ·
11 vector spins
spreading: 03 contagion · 08 copying · 09 Hawkes ·
13 cultural conventions
information: 04 semantic information · 12 info dynamics
replicators & thermo: 05 · 06 · 07 · 15 stochastic thermo
fluctuations: 14 scaling and Taylor laws
each folder: model · village mapping · fit · nulls · pitfalls ·
“what held up” after round 1
DEFINITIONS.md: 393 operational terms and variants

catalogs
goal-periods.md: 51 periods with N, regime, mode, models
natural-experiments.md: **NE01–NE46**, dated step
changes from the CHANGELOG and logs = our interventions

locked holdout holdout.md + holdout_ledger.json
periods and NE windows reserved since 2026-10-03; masked in
every exploratory script; 46 ledger items (uses, disclosures);
held-out labels stored apart, unexamined until confirmation

3 · Ideas → hypotheses → scores

HYPOTHYPOTHESES.md HH001–HH375, append-only
loose ideas tagged by model, period and the four impostors; batches
go to the dashboard vetting strip; approved ones get an H number

hypotheses/H01–H132/ one research direction each
card: question (Q1–Q7) · model · data scheme · null ·
impostor table · **dated predictions and kill rules, before data**
round 1: synthetic run on the real skeleton → dated amendments
→ two layers: replication + period-native tests (NEs)
2,805 period folders (GNN / NENN), each with verdict and role
confirm.py: frozen by hash, guarded, dry-run only
2-page RevTeX summary · estimates rows · “claim that stands”

status
round 1 done for 110+ hypotheses; round 2 for H01–H39, H81, H99;
newest batches H106–H132 (magnet, kinetic Ising) running;
holdout runs so far: H02, H04, H05; the rest await sign-off

scoring v2 (page 2) **summary/meta.json**
credence: base rate × evidence factors, capped at 0.95 ·
coordinator + blind rater, log-odds mean · adjudication rules 1–9 ·
value V , $EU = pV$, mechanism M0–M2, fragile flag ·
faithfulness scorecard A–I (0/1/2) and promotion levels
112 claims scored; 92 multi-rater; median credence 0.66

how the work runs
one coordinator session + up to 20 parallel agents (one or two
hypotheses each); agents never edit shared files; the coordinator
integrates each report: score, constants, issues, ledger, LOG, commit

next first confirmatory holdout runs (H08, H40 on NE20,
H67, H111) after Vivian's sign-off; round 2 for the strongest leads

4 · Outputs

interpretation/
swarm-constants.json: 71
constants with scope, source
lever-table.md: operator levers
phase diagram: $N \times \text{regime}$, g_{lag}

writeup/ (RevTeX)
paper.tex · compendium (2 pp.
per hypothesis) · syntheses ·
operator guide (47 instructions)
visuals/: figures + animations

dashboard/ (local)
model × hypothesis matrix,
sortable by credence and EU
phase diagram · constants ·
HH vetting strip

OVERVIEW.md
generated hypothesis × period
verdict table (never hand-edited)

answers to Q1–Q7 (GOALS.md)
coupling = reading at the next call;
fields beat coupling; subcritical;
one collective slow mode (open)

Runs through everything: **CLAUDE.md** map & conventions · **GOALS.md** Q1–Q7 · **STANDARDS.md** four impostors (scheduler field, exogenous field, shared priors, contemporaneous convergence), partition contrasts, two layers · **LOG.md** lab notebook · **_provenance.json** on every processed folder · **predictions before data** · **one goal period = one unit** (compare fitted parameters, never pool) · compute ≤ 2 workers per agent · storage ≤ 20 GB · data terms (no text in commits; cite AI Digest)

How a claim is scored: credence and faithfulness

Credence: will this claim survive an independent confirmatory test? Faithfulness: does the model capture the mechanism, not just the statistics? Green letters link each factor to an axis.

(a) Credence ladder (scoring v2, one per hypothesis)

1 • What is scored one clause: the card’s “**claim that stands**”, with its scope (periods, units, effect size). Withdrawn, post hoc or unpowered sub-results are named exclusions. A negative claim states the effect size at which its power reaches 0.8 and is scoped to it; below that it is “inconclusive”.

2 • Base rate p_0 from provenance

0.40 pre-registered primary prediction and estimator fixed before data

0.30 secondary estimator redesigned after data

0.20 post hoc claim found in the data

3 • Multiply the odds $p/(1 - p)$ by each factor that applies (one design is credited once)

C calibrated null passed (DQ8 sizes or own synthetic)	×2	(mis-sized ×0.7)
I replication: same sign in ≥ 2/3 of ≥ 6 independent units	×2	(one unit ×0.6)
B robust to data versions, embedding models, estimators	×2	(changed by a fix ×0.3)
F identified: synthetic recovery at the real counts	×1.5	(not identifiable ×0.5)
E consistent with a natural experiment (intervention)	×2	
H rival model beaten on held-out data	×1.5	
G ground truth agrees (DQ6 roles, rooms, scheduler)	×1.5	
I locked holdout passed / failed	×5 / ×0.15	
penalties: unvalidated LLM labels, problem inputs, multiplicity	×0.5–0.8 each	

4 • Credence $p = \text{odds}/(1 + \text{odds})$, capped at **0.95** before a holdout run (0.98 after)

5 • Two raters
coordinator + a **blind** rater (never sees the other score) → log-odds mean.
 $|\Delta p| > 0.25$ or $|\Delta V| > 1$ → adjudication under rules 1–9 (score one clause; scope negatives by power; same-data agreement is robustness, not replication; ...)

6 • Use and depth
value if true V (0–5): the decision it changes, magnitude in that decision’s units, generality
 $EU = pV$ (dashboard sorts by this)
M0 pattern · **M1** signature + rival rejected · **M2** intervention + rival + synthetic
fragile: headline moved across versions

Worked example: H111 (talk obeys the Fano sum rule in regime III)

p_0
0.40

→ unfitted prediction from H67’s g_{lag} ×2
0.57

→ 14/18 units ×1.5
0.67

→ synthetic recovery ×1.3
0.72

→ g from the same data ×0.9
0.70

→ NE42 native failed ×0.8
0.65

⇒ $p = \mathbf{0.65}$
 $V = 2, EU\ 1.3, M1$

Rules: writeup/hypothesis-pages/RUBRIC.md (scoring v2, adjudication rules 1–9) · STANDARDS.md (impostors, two layers) · paper.tex §Assessing model faithfulness. State as of 2026-10-04.

(b) Faithfulness scorecard (per model, mapping and window)

Fit is not faithfulness. A maximum-entropy model reproduces its fitted correlations by construction. Each axis scores ○ 0 not done or failed, ◐ 1 partial, ● 2 passed, with evidence linked from the card.

A mapping
every variable defined from dataset fields; assumptions listed; invariant across model families and regimes

B assumptions
stationarity (split halves), Markov order, Hawkes time-rescaling, update-order audit; robust across data versions

C adequacy
beats the null hierarchy on held-out data: the **four impostors** (scheduler field, exogenous field, priors, convergence)

D unfitted predictions
reproduces statistics left out of the fit and the model’s signature (a jump, $s^{-3/2}$ sizes, a sum rule)

E interventional
fitted on one side of a natural experiment, predicts the sign and size on the other side, or across a reversal

F identifiability
synthetic recovery with the village’s real sampling; stable over bins, embedding model, split halves

G ground truth
agrees with known structure: assigned roles and leaders, rooms, families, scheduler changes

H comparative
beats the rival models named on the card (held-out likelihood, description length, partition contrasts)

I transfer
holds in other periods of the same mode and regime, **including the locked holdout**

Promotion levels (each step needs the one before)

idea → hypothesis
 $A \geq 1$ + written plan for C, D and E or G, with falsifiers

→ **descriptive**
 $C = 2, D \geq 1$

→ **supported**
 $A = 2, B \geq 1, C = D = 2, F \geq 1, H \geq 1, E$ or $G = 2$, on the holdout

→ **faithful**
 $\text{supported} + E = F = H = I = 2$ (mechanistic)

Example scorecard: H111

A B C D E F G H I

scores 1 1 1 2 1 1 1 1 0: an unfitted prediction passed ($D = 2$), but C needs a held-out null win and I needs the holdout. Level: **hypothesis**. Credence 0.65 says the claim is likely to hold; it does not say the model is the mechanism.

